

BSG - Statistical Genetics - practical 1

15/09/2025, submission deadline 21/09/2025 (midnight)

Resolve the following exercise in groups of 3-4 students. Write the R scripts, perform the computations and make the graphics that are asked for in the practical below. Take care to give each graph a title, and clearly label x and y axes, and to answer all questions asked. You can write your solution in a Word or Latex document or generate a html/pdf file with R Markdown. Take care to number your answers exactly as in this exercise. Upload your solution the web page of the course at raco.fib.upc.edu no later than the submission deadline.

You can make use of the R-package `genetics` (and other packages) to compute your answers, as you please. The datasets can be downloaded from the web page of the course at raco.fib.upc.edu.

The first part of the practical is dedicated to the descriptive analysis of SNP data, whereas the second part is dedicated to the analysis of STR data.

SNP dataset

The file `TSICHR22RAW.raw` contains genotype information of individuals from Tuscany in Italy, taken from the 1,000 Genomes project. The datafile contains all single nucleotide polymorphisms on chromosome 22 for which complete information is available.

Load this data into the R environment, with the `read.table` instruction. The first six columns contain non-genetical information. Create a dataframe that only contains the genetic information that is in and beyond the 7th column. Notice that the genetic variants are identified by an “rs” identifier. The genetic data is coded in the (0, 1, 2) format with 0=AA, 1=AB, 2=BB.

1. How many variants are there in this database? What percentage of the data is missing?
2. Calculate the percentage of monomorphic variants. Exclude all monomorphics from the database for all posterior computations of the practical. How many variants do remain in your database?
3. Report the genotype counts and the minor allele count of polymorphism rs8138488-C, and calculate the MAF of this variant.
4. Compute the minor allele frequencies (MAF) for all markers, and make a histogram of it. Does the MAF follow a uniform distribution? What percentage of the markers have a MAF below 0.05? And below 0.01? Can you explain the observed pattern?

Uniform distribution at $[0, 0.5]$. The range from 0 to 0.5 can be easily explained due to the fact that a frequency higher than 0.5 would indicate this is not the minor allele.

5. Calculate the observed heterozygosity H_0 , and make a histogram of it. What is, theoretically, the range of variation of this statistic?
6. Compute for each marker its expected heterozygosity (H_e), where the expected heterozygosity for a bi-allelic marker is defined as $1 - \sum_{i=1}^k p_i^2$, where p_i^2 is the frequency of the i th allele. Make a histogram of the expected heterozygosity. What is, theoretically, the range of variation of this statistic? What is the average of H_e for this database?

STR dataset

The object `NistSTRs` of the R package `HardyWeinberg` contains a set of STRs of individuals of Caucasian ancestry, which can be loaded with the instructions `library(HardyWeinberg)` and `data(NistSTRs)`. The rownames of the object consist of identifiers for each individual. Successive columns represent the two alleles of an individual for each STR. Note there exist fractional alleles (like 14.3) that indicate the particular STR sequence is repeated in-between a certain numbers of

times. These fractional alleles are regarded as separate alleles (e.g. 14.3 is different from 14 and 15).

1. How many individuals and how many STRs contains the database?
2. Write a function that determines the number of alleles for a STR. Determine the number of alleles for each STR in the database. Compute basic descriptive statistics of the number of alleles (mean, standard deviation, median, minimum, maximum).
3. Make a table with the number of STRs for a given number of alleles and present a barplot of the number STRs in each category. What is the most common number of alleles for an STR?
4. Compute the expected heterozygosity for each STR. Make a histogram of the expected heterozygosity over all STRS. Compute the average expected heterozygosity over all STRs.
5. Calculate also the observed heterozygosity for each STR. Plot observed against expected heterozygosity, using all STRs. What do you observe?
6. Compare, overall, the results you obtained for the SNP database with those you obtained for the STR database. What differences do you observe between these two types of genetic markers?